CSCI 4311 · Mobile Application Development

Week 6 Lab — Campus Buddy KICT: The App Shell

Time: ~90 min (after TEST 1) · **You need:** Week 5 checkpoint

Goal: the app gets more than one *area*: bottom buttons (Home / Courses / Profile), a detail page for one course, and an About page.

> ✓ = *what you should see.* ★ *CHECKPOINT = show your demonstrator.*

A. Bottom navigation (25 min)

Step 1. First, tidy the project — this shape lasts until Week 14:

1. Make a folder `lib/screens/`. 2. Move `profile_screen.dart` and `courses_screen.dart` into it. (Drag in your editor, or move the file and let the editor fix imports — if it doesn't, fix the red errors by hand.) 3. Make a new file `lib/screens/home_screen.dart` — for now just:

```
import 'package:flutter/material.dart';

class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Campus Buddy KICT')),
      body: const Center(child: Text('Home — coming in Week 7')),
    );
  }
}
```

Step 2. Make `lib/shell_screen.dart`:

```
import 'package:flutter/material.dart';
import 'package:campus_buddy_kict/screens/home_screen.dart';
import 'package:campus_buddy_kict/screens/courses_screen.dart';
import 'package:campus_buddy_kict/screens/profile_screen.dart';

class ShellScreen extends StatefulWidget {
  const ShellScreen({super.key});

  @override
  State<ShellScreen> createState() => _ShellScreenState();
}

class _ShellScreenState extends State<ShellScreen> {
  int _tab = 0;

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: IndexedStack(
```

```
        index: _tab,
        children: const [
          HomeScreen(),
          CoursesScreen(),
          ProfileScreen(),
        ],
      ),
      bottomNavigationBar: NavigationBar(
        selectedIndex: _tab,
        onDestinationSelected: (i) => setState(() => _tab = i),
        destinations: const [
          NavigationDestination(icon: Icon(Icons.home_outlined),
              selectedIcon: Icon(Icons.home), label: 'Home'),
          NavigationDestination(icon: Icon(Icons.list_outlined),
              selectedIcon: Icon(Icons.list), label: 'Courses'),
          NavigationDestination(icon: Icon(Icons.person_outline),
              selectedIcon: Icon(Icons.person), label: 'Profile'),
        ],
      ),
    );
  }
}
```

> *Tip:* `NavigationBar` *= the bottom bar. Tapping a destination changes* `_tab`*. Tip:* `IndexedStack` *=* ***keep all three screens alive****, show one at a time. That's why a starred course is still starred when you come back.*

Step 3. In `main.dart`: import the shell, `home: const ShellScreen()`. Press `R`.

★ CHECKPOINT 1 — three buttons at the bottom switch screens. Star a course, switch to Home, come back — star still there.

B. Course detail page (35 min)

Step 4. Make `lib/screens/course_detail_screen.dart`:

```
import 'package:flutter/material.dart';
import 'package:campus_buddy_kict/models/course.dart';

class CourseDetailScreen extends StatelessWidget {
  const CourseDetailScreen({super.key, required this.course});

  final Course course;

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text(course.code)),
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
```

```
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            Row(children: [
              Icon(course.icon, size: 48,
                  color: Theme.of(context).colorScheme.primary),
              const SizedBox(width: 16),
              Expanded(
                child: Text(course.name,
                    style: Theme.of(context).textTheme.headlineSmall),
              ),
            ]),
            const SizedBox(height: 16),
            Text('${course.creditHours} credit hours · ${course.day}'),
            const SizedBox(height: 24),
            const Spacer(),
            SizedBox(
              width: double.infinity,
              child: OutlinedButton.icon(
                icon: const Icon(Icons.delete_outline),
                label: const Text('Remove this course'),
                onPressed: () => Navigator.pop(context, true),  // the answer
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

> *Tip: Two ideas, both reusable forever:*
>
> – Data goes **in** through the constructor: `required this.course`. A screen is just a widget — give it arguments like any widget.
>
> – An answer comes **back** with `Navigator.pop(context, true)` — the same trick the Week 5 popup used.

Step 5. In `courses_screen.dart`, make a row tap open the detail page. Replace the row's `onTap:` with:

```
onTap: () async {
  final removed = await Navigator.push<bool>(
    context,
    MaterialPageRoute(
      builder: (_) => CourseDetailScreen(course: _courses[index]),
    ),
  );
  if (removed == true) {
    setState(() => _courses.removeAt(index));
  }
}
```

> *Tip:* `Navigator.push` *slides a new page on top and* **waits** *(*`await`*) until it closes. Closed with* `pop(true)` *→ you get* `true`*. Closed with the back button → you get* `null`*. That's why the* `if` *checks* `== true`*.*

Press `r`.

★ CHECKPOINT 2 — tap a row: detail page slides in, with a ← back button you never wrote. Press "Remove this course" → back on the list, row gone, totals updated.

Try it yourself (5 min)

Replace `IndexedStack` with `children: [ if (_tab == 0) HomeScreen(), if (_tab == 1) CoursesScreen(), if (_tab == 2) ProfileScreen(), ]`. Star a course, switch tabs, come back. What died? Put `IndexedStack` back. Now you know what it's for.

C. About page with a named route (15 min)

There's a second way to open pages: by **name**, like an address.

Step 6. In `main.dart`, inside `MaterialApp(`, add:

```
routes: {
  '/about': (context) => const AboutScreen(),
},
```

Step 7. Make `lib/screens/about_screen.dart`: a Scaffold with the course name, your name as developer, `1.0.0 · Week 6`, and a TextButton "Close" that runs `Navigator.pop(context)`.

Step 8. In Home's AppBar, add an info icon that opens it:

```
IconButton(
  icon: const Icon(Icons.info_outline),
  onPressed: () => Navigator.pushNamed(context, '/about'),
)
```

★ CHECKPOINT 3 — About opens from Home; the system back gesture also closes it.

> *Tip: When to use which?* `push` *+ constructor = when the page needs data (like one course). Named route = for fixed pages like About, openable from anywhere.*

D. Home dashboard (15 min)

Step 9. Replace Home's body with a menu of cards:

```
ListView(
  padding: const EdgeInsets.all(16),
  children: const [
```

```
    Card(child: ListTile(
      leading: Icon(Icons.wb_twilight),
      title: Text('Prayer times'),
      subtitle: Text('Arrives in Week 8'),
      trailing: Icon(Icons.chevron_right),
    )),
    Card(child: ListTile(
      leading: Icon(Icons.link),
      title: Text('Campus links'),
      subtitle: Text('Arrives in Week 7'),
      trailing: Icon(Icons.chevron_right),
    )),
    Card(child: ListTile(
      leading: Icon(Icons.info_outline),
      title: Text('About Campus Buddy KICT'),
      trailing: Icon(Icons.chevron_right),
    )),
  ],
)
```

Step 10. Make each card do something: About → `pushNamed('/about')`. Prayer times and Campus links → push a tiny placeholder screen each (make two 10-line stubs — they become real in Weeks 7 and 8).

★ CHECKPOINT 4 — Home is a menu; all three cards go somewhere.

E. Save your week (3 min)

```
git add .
git commit -m "week 6 app shell"
```

`.gitignore` **this week: nothing to change.** Moving files into `screens/` shows in `git status` as deletes + adds — that's correct, let git save it.

Exercise in class (hand in before leaving)

A screen that answers, part 2. On Home, add a "clear all favourites" button. It must first open a whole confirmation **screen** (not a popup — a Scaffold with "Are you sure?" + Yes/No buttons) that pops `true` or `false`. On `true`… you can't actually clear anything: the favourites live inside the Courses screen, and Home can't reach them. Write 3–4 sentences explaining why not, and name the concept that would fix it (state management — we'll use it, not teach it, this semester). Make the button work when pressed from the Courses tab instead.

Homework (Assignment 4, graded)

1. Add a **fourth tab**: `Notes` (a stub screen — Notes arrive Week 11). 2. Give the detail page an **Edit** pencil icon. It opens `EditCourseScreen(course)` with the form pre-filled. Saving pops the edited `Course` object back, and the list replaces that row (`copyWith` from Week 5, `setState`). 3. Protect it: if the user presses the system back from the edit screen, nothing changes (back = `null` = cancelled — check before using the result, like Step 5 above).

If something goes wrong

Problem	Fix
Screen went blank after moving files	Import paths — check every `package:campus_buddy_kict/...` line
Two bars stacked on top of each other	Each screen has its own Scaffold; the shell's Scaffold has no AppBar on purpose
Wrong row disappears after Remove	Capture the course, not the position: `onTap` gives you `_courses[index]` — keep it in a variable before `await`, then `removeAt(_courses.indexOf(course))`
"Navigator context error"	`Navigator` must be called from inside `build`'s widgets, not from `main.dart`
`onGenerateRoute` error on About	Route name typo — `'/about'` must match exactly

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Next lab:

the Campus links page goes live — real web links, the IIUM map, one-tap email and phone, sharing. Your first package from pub.dev.